

MATERIAL SAFETY DATA SHEET

SECTION 1. CHEMICAL IDENTIFICATION

Product name:	Kasugamycin 3%+Tricyclazole 10% WP
Company Name:	Ningbo Greenidea Chemicals Co., Ltd.
Common Name of A.I.:	Kasugamycin or <i>Kasugamycin hydrochloride hydrate</i> , and Tricyclazole
Chemical Name:	Kasugamycin: L-1,3,4/2,5,6-1-deoxy-2,3,4,5,6-pentahydroxycyclohexyl 2-amino-2,3,4,6-tetradeoxy-4-(α -iminoglycino)- α -D- <i>arabino</i> -hexopyranoside hydrochloride hydrate; [5-amino-2-methyl-6-(2,3,4,5,6-pentahydroxycy- clohexyloxy) tetrahydropyran-3-yl]amino- α - iminoacetic acid hydrochloride hydrate Tricyclazole: 5-methyl-1,2,4-triazolo[3,4- <i>b</i>][1,3]benzothiazole
Synonym:	None
Chemical Family:	Kasugamycin: Hexopyranosyl antibiotic Tricyclazole: MBI: reductase
Chemical Formulation:	Kasugamycin: C ₁₄ H ₂₅ N ₃ O ₉ Tricyclazole: C ₉ H ₇ N ₃ S
Formation:	Wettable Powders

SECTION 2. COMPOSITION/INFORMATION ON INGREDIENTS

Component Name	CAS No.	Content (w/w)
Kasugamycin	6980-18-3	30 g/kg
Tricyclazole	41814-78-2	100g/kg
Other ingredients	/	Fill to 1kg

Ingredients not precisely identified are proprietary or non-hazardous.

SECTION 3. HAZARDS IDENTIFICATION

Hazard Classification:

Hazardous. Dangerous for the environment.

Acute Health Effects:

Swallowed: Harmful if swallowed.
Eye: Will irritate eyes. May cause weeping and redness.
Skin: Will irritate skin. It does not readily penetrate the skin.
Inhaled: Harmful if inhaled.

Risk phrases:

R22 Harmful if swallowed.

Safety phrases:

S2: Keep out of the reach of children

S3: Keep in a cool place

S13: Keep away from food, drink and animal foodstuffs

S15: Keep away from heat

S20/21: When using do not eat, drink or smoke

S35 This material and its container must be disposed of in a safe way.

S57 Use appropriate container to avoid environmental contamination.

SECTION 4. FIRST AID MEASURES

First Aid Measures

Ingestion: If swallowed: Call a poison control center or doctor immediately for treatment advice. Have the person sip a glass of water if able to swallow. Do not induce vomiting unless told to do so by a poison control center or doctor.

Eye Contact: If in eyes: Hold eye open and rinse slowly and gently with water for 15-20 minutes. Remove contact lenses, if present, after 5 minutes, then continue rinsing eye. Call a poison control center or doctor for treatment advice.

Skin Contact: If on skin or clothing: Take off contaminated clothing. Rinse skin immediately with plenty of water for 15-20 minutes. Call a poison control center or doctor for treatment advice.

Inhalation: If inhaled: Move person to fresh air. If person is not breathing, call an ambulance, then give artificial respiration, preferably mouth-to-mouth if possible. Call a poison control center or doctor for further treatment advice.

Note to Physician

Treat symptomatically. There is no specific antidote.

SECTION 5. FIRE FIGHTING MEASURES

Fire and Explosion

Flash Point: Not available

Autoignition Temperature: Not available

Flammability: Not flammable

Unusual Fire, Explosion and Reactivity Hazards

During a fire, irritating and possibly toxic gases may be generated by thermal decomposition or combustion.

In Case of Fire

Use dry chemical, foam or CO₂ extinguishing media. Wear full protective clothing and self-contained breathing apparatus. Evacuate nonessential personnel from the area to prevent human exposure to fire, smoke, fumes or products of combustion. Prevent use of contaminated buildings, area, and equipment until decontaminated. Water runoff can cause environmental damage. If water is used to fight fire, dike and collect runoff.

SECTION 6. ACCIDENTAL RELEASE MEASURES

Emergency Procedures

In case of spillage it is important to take all steps necessary to:

- Avoid eye and skin contact.
- Avoid contamination of waterways

Methods and Materials for Containment and Clean Up

Procedure for spill:

- (1) Keep all bystanders away.
- (2) Wear goggles, half face-piece respirator with combined dust and vapour cartridge, full length clothing and PVC gloves.
- (3) Reposition any leaking containers so as to minimise further leakage.
- (4) Dam and absorb spill with an absorbent material (e.g. sand or soil).
- (5) Shovel the absorbed spill into drums.
- (6) Disposal of the absorbed material will depend upon the extent of the spill.
 - For quantities up to 50L of product bury in a secure landfill site.
 - For quantities greater than 50L seek advice (use emergency contact number below) from the manufacturer before attempting disposal. Contain in a secure location until disposal method is established.
- (7) Decontaminate the spill area with detergent and water and rinse with water

SECTION 7. HANDLING AND STORAGE

Store the material in a well-ventilated, secure area out of reach of children and

domestic animals. Do not store food, beverages or tobacco products in the storage area. Prevent eating, drinking, tobacco use, and cosmetic application in areas where there is a potential for exposure to the material. Wash thoroughly with soap and water after handling.

SECTION 8. EXPOSURE CONTROLS/ PERSONAL PROTECTION

Engineering Controls:

Control process conditions to avoid contact. Use local exhaust ventilation during manufacture. Use in a well-ventilated area only.

Personal Protection:

Eyes: Safety goggles

Clothing: Cotton overalls buttoned to the neck and wrist (or equivalent clothing)

Gloves: Elbow-length PVC gloves

Respiratory: Do not breathe vapour or spray. Wear an organic cartridge respirator if inhalation is possible.

Other: Wash hands after use. After each day's use, wash gloves, goggles, respirator facepiece and contaminated clothing.

SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance:	Light Brown Color Powder
Odor:	Special odor
Boiling Point/Melting Point:	202-204°C (Kasugamycin) 275°C (Tricyclazole)
Vapour Pressure:	$<1.3 \times 10^{-2}$ mPa (25°C) (Kasugamycin) 5.86×10^{-4} mPa (20°C) (Tricyclazole)
Flashpoint:	Non-flammable
Flammability Limits:	Not applicable
Solubility in Water:	Suspend in water
pH:	5.0 – 7.0
Formulation:	Wettable Powder

SECTION 10. STABILITY AND REACTIVITY

Reactivity Hazards: Stable under normal conditions of use. No dangerous reaction known under normal conditions of use. Avoid extreme heat and fire. Toxic thermal decomposition products may include oxides of carbon, sulphur and nitrogen, and

compounds of chlorine and fluorine.

SECTION 11. TOXICOLOGICAL INFORMATION

Acute Oral Toxicity:	Rat: LD ₅₀ : >2000 mg/kg.
Acute Dermal Toxicity:	Rabbit: LD ₅₀ : > 2000 mg/kg.
Acute Inhalation Toxicity:	Rat: LC ₅₀ (4 h): 1.36 mg/L.
Skin Irritation:	Rabbit: Practically non-irritating.
Eye Irritation:	Rabbit: Non eye irritating.
Sensitization:	Guinea pig: Not a sensitizer.

SECTION 12. ECOLOGICAL INFORMATION

For Kasugamycin technical:

Birds: Acute oral LD₅₀ for male Japanese quail >4000 mg/kg.

Fish: LC₅₀ (48 h) for carp and goldfish >40 mg/l.

Daphnia: LC₅₀ (6 h) >40 mg/l.

Bees: LD₅₀ (contact) >40 µg/bee.

For Tricyclazole technical:

Birds: Acute oral LD₅₀ for mallard ducks and bobwhite quail >100 mg/kg.

Fish: LC₅₀ (96 h) for bluegill sunfish 16.0, rainbow trout 7.3, goldfish fingerlings 13.5 mg/l.

Daphnia: LC₅₀ (48 h) >20 mg/l; NOEC (21 d) 0.96 mg/l.

Bees: Acute contact LD₅₀ in honey bee (*Apis mellifera*) is >100 µg /bee, and acute oral LD₅₀ > 88.5 µg/bee.

Algae: Growth inhibition EC₅₀ in green alga (*Selenastrum capricornutum*) 9.0 mg/L.

Earthworms : The LC₅₀ in earthworm (*Eisenia foetida*) >1000 mg/kg.

SECTION 13. DISPOSAL CONSIDERATIONS

Waste Disposal Methods:

Triple or preferably pressure rinse containers before disposal. Add rinsings to spray tank. Do not dispose of undiluted chemicals on site. Break, crush, or puncture and bury empty containers in a local authority landfill. If no landfill is available, bury the containers below 500 mm in a disposal pit specifically marked and set up for the purpose, clear of waterways, desirable vegetation and tree roots. Empty containers and product should not be burnt.

SECTION 14. TRANSPORT INFORMATION

Land transport ADR/RID

UN number: Not regulated

Proper Shipping Name: Not regulated

Class: Not regulated

Subsidiary Risk: Not regulated

Packaging group: Not regulated

Marine transport IMDG

UN number: 3077

Proper Shipping Name: Environmental hazardous substance, liquid, n.o.s. (contains Kasugaymycin, Tricyclazole)

Class: 9

Subsidiary Risk: Not regulated

Packaging group: III

Marine pollutant: Yes

Air transport ICAO/IATA

UN number: 3077

Proper Shipping Name: Environmental hazardous substance, liquid, n.o.s. (contains Kasugaymycin, Tricyclazole)

Class: 9

Subsidiary Risk: Not regulated

Packaging group: III

SECTION 15. REGULATORY INFORMATION

Registered according to the Agricultural and Veterinary Chemicals Act 1988

See also section 2.

SECTION 16. OTHER INFORMATION

Revised Date: January 15, 2016

Revision Summary:

To reflect database modification (no substantive changes - only related to system)

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